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Bachelor Thesis

”Machine Learning-Based Predictive Modeling of Energy Prices”

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SUMMARY

This project focuses on the development of a machine learning-based predictive model for electricity prices in Spain. Using historical data from the OMIE and technical analysis (TA) indicators, the model aims to accurately forecast hourly energy prices. The study focuses on a single hourly slot, evaluating the performance of various Machine Learning models, such as linear regression, Lasso, and Random Forest. There was a strong focus on the feature engineering, employing technical analysis indicators such as moving averages, exponential moving averages, and momentum metrics. The results display the impact of tailoring the features to improve model accuracy and offer insights into the potential of data-driven approaches for energy price forecasting.

Keywords:

Energy

Machine Learning

Predictive Modeling

Technical Analysis (TA)

DEDICATION

Dedicated to my late grandfather who was not able to see me become an engineer like himself
Here is to my amazing family
To my parents, that have always To my grandparents
To my girlfriend
To my friends and colleagues
To anyone and everyone that has supported me during my years in university

Here is to the next steps in life

CONTENTS

LIST OF FIGURES

LIST OF TABLES

1. INTRODUCTION TO THE PROBLEM

- **1. Explanation of how the price of energy works in Spain**
  - a. Energy
  - b. Prices
- **2. Machine Learning - Lets run through some model overviews, which are available to choose, but without actually referencing anything**

Energy has increasingly become more important. Availability is key for a modern high functioning society.q In Europe we are lucky to have a really advanced and reliable network. It is a key resource needed to succeed in the ecological transition that we are currently undergoing world wide. Electric mobility is a key element of this process. Electric cars, buses and trucks, need easily available and convenient sources of energy to recharge, to be able to play their role in society. An important factor in this ever growing necessity is the price of the available energy. But in recent times, that easy and affordable access has changed, and has become more unpredictable. Ever since the commencement of the war between Russia and Ukraine marked and inflection point in the mostly cyclical nature of electricity prices. Prices have gone up considerable since a couple years time, and it has been affecting everyone, from big consumers such as companies, to everyday people, having to pay a more expensive electricity bill at the end of the month. This has become a topic of great concern, with more and more people thinking about it often.

This led me to think about pricing, how it was structured, and what to expect as an end consumer. Could we plan our use of energy according to the price? Could it be predicted accurately? These were some of the questions I had that sparked my interest in looking into energy predictions.

In the initial discussions of this project, we talked about various ways to attack the problem at hand. We thought about a variety of systems but settled on a plan to explore the predictions with the more established methods of Machine Learning. This was done specially since it was a natural continuation of what was learned in the third year subject of Telematics Engineering, Modern Theory of Detection and Estimation.

The first idea we developed was to separate out the project in 24 blocks, one for each hour block to predict. The reason behind splitting the data in 24 blocks was simple, we assumed that data between the same time slot across neighbor days, would be similar, or would follow a certain trend.

Focusing on the innovation side of things, we decided to simplify the problem. Pivoting from creating an algorithm to predict hourly prices, to splitting that into the 24 blocks, and predicting a single one. We decided on the 14th hour of each day, since irrespective of the day of the week, or holiday, it would always be a valley period.

This kind of granular breaking up of data is similar to what a Random Forest does in order to absorb more details and create higher resolution predictions. We made this assumption based on the idea that it would simplify the project substantially.

Talk about Emilio's background in investment banking at BBVA and his alorithms.

Talk about the idea of using technical analysis for feature engineering

2. BACKGROUND AND RELATED WORK

- **References**
  - How is the price of electricity / energy set? What happens if new producers enter the market?
  - Articles that reference energy prices prediction as an inspiration for my own work:
    * 5-10 articles that do/talk about similar things
    * Energy consumption predictions
    * Energy prices in other European countries (Italy, Germany, UK, France)
    * Longer term predictions
    * Predicciones de produccion
  - Describir tecnologias – explicar mas en detalle como funciona y enlaces a donde pueda averiguar el contenido
- Referenciar que metricas uso para medir mi error etc
- Referencias de donde saco los datos
- Referencias a que software utilizo referencias a documentacion de sklearn etc

How is the price of electricity set in Spain and Portugal?

What happens when new producers enter the market and sell to the energy pool?

Comment about more typical case studies focusing on energy consumption, since that is something more consistent and predictable

2.1. The Sliding Window plan: Setting up the Feature Matrix

How do I prepare my data to utilize across different models: The Feature Matrix (weight matrix)

The sliding window theory (mini-models)

2.2. Which Machine Learning models will be used

What algorithms will I use?

- Linear Regression

- Lasso Regression

- Random Forest

2.3. Fine Tuning with Technical Analysis and Feature engineering

What Type of finetuning will I use?

- Feature engineering & selection

- Technical analysis based feature engineering

TA metrics used? SMA, EMA, ROC & RSI

Talk about the intervals used:

$SMA_3, SMA_5, SMA_7, SMA_{14}, SMA_{30}, SMA_{60}, SMA_{90}, SMA_{180}, SMA_{360}$

$EMA_3, EMA_5, EMA_7, EMA_{14}, EMA_{30}$

$ROC_3, ROC_5, ROC_7, ROC_{12}, ROC_{14}, ROC_{30}$

RSI_5, RSI_7, RSI_{14}

2.4. Validation of results

How will I measure my accuracy? errors: R^2 RMSE MSE

3. THEORETICAL SYSTEM DESCRIPTION: PROPOSAL

This chapter should be the complete pipeline explanation to my boss

This chapter will outline the theoretical design of the system developed for the project. It will outline the complete pipeline constructed for the system: starting from the data preparation stage, continuing with the data preparation to the final results.

Data Preparation

Matrix Creation

Mini-models - sliding window approach

Technical indicators

ML blocks Linear Regression Linear combination of every feature Lasso Regression Can turn unimportant features into 0s Random Forest Feature engineering

Pandas, NumPy, scikit-learn

Retrieve Data from OMIE

Ingest data and create a CSV db

Create Sliding Window Matrix for use in training that batch

Train the batch

Cross validation?? Not really done

Review the mini model theory, why is it reasonable to not use training set, validation and test sets?

Review error rates and compare with other iterations/ batches and other models.

(make a block diagram of all this)

4. EXPERIMENTATION

4.1. Data Description – OMIE data

How is the data provided?

4.2. Set Up Explanation

How do I use the previously explained system - what parameters did I modify to obtain the final results. Feature engineering, Alpha variation testing in Lasso, Tree depth and number of leafs.

4.3. Results of the testing

Results as graph/ tables - This would be to compare a model across various iterations. Model best-run comparisons, etc.

4.4. Low level discussion

This would be the final explanation that I would give to a colleague of mine, with full details on how I have done everything

1. Download of OMIE data [1]

Downloaded with a Python script for the dates that were not available with an easy.zip file (for whole previous years)

Insert part of the python script?? or better in point 4

2. The data has this shape:

```
MARGINALPDBC;
2018;01;01;1;28.1;6.74;
2018;01;01;2;33;4.74;
2018;01;01;3;32.9;3.66;
2018;01;01;4;28.1;2.3;
```

```
2018;01;01;5;27.6;2.3;
2018;01;01;6;24.6;2.06;
2018;01;01;7;20.1;2.06;
2018;01;01;8;19.9;2.06;
2018;01;01;9;19.84;2.3;
2018;01;01;10;19.9;2.3;
2018;01;01;11;19.9;2.3;
2018;01;01;12;19.9;2.3;
2018;01;01;13;23.6;2.3;
2018;01;01;14;25.1;2.3;
2018;01;01;15;23.6;5;
2018;01;01;16;24.6;5;
2018;01;01;17;25.1;5;
2018;01;01;18;27.6;8.85;
2018;01;01;19;27.6;15.93;
2018;01;01;20;28.1;22.02;
2018;01;01;21;32.9;20;
2018;01;01;22;28.1;21.95;
2018;01;01;23;28.1;23.52;
2018;01;01;24;27.6;16.35;
*
```

3. Data format: In page number 67, chapter 6.18 we may find the following information regarding the encoding format for the information: [2]

```
6.18 Precios marginales del mercado diario (MARGINALPDBC)

Fichero con los precios marginales del Mercado Diario para cada una
de las horas.
Nombre del fichero: marginalpdbc_aaaammdd.v donde aaaammdd
corresponde a la fecha de sesión y v es la versión del fichero.

Descripción de los campos:
CAMPO DESCRIPCIÓN VALORES VÁLIDOS
Año Año I4 - 20XX
Mes Mes I2 - 1 a 12
Día Día I2 - 1 a 31
Hora Hora I2 - 1 a 25
MarginalPT Precio marginal zona Portuguesa F8.2 -
-99999.99 a 99999.99
MarginalES Precio marginal zona Española F8.2 -
-99999.99 a 99999.99
```

4. Data organization

We will transform the data into a CSV format easier to use with the Pandas library to later create a DataFrame. We will maintain the same information, but in a encoding more suitable for Pandas. For that, the date and time must be transformed to a format such as Datetime which will transform the information from `2018;01;01;14; ...` to `2018-01-01 14:00:00`

5. Data visualization with Seaborn? With Microsoft Corporation's "Data Wrangler" VSCode extension

6. Data retention of only the relevant info time and date, and MarginalES - drop MarginalPT

7. Data cleanup: Check for missing or duplicate entries. since Daytime Savings introduces duplicate hours some days of the year: INSERT some dates in the dataset that have the duplicated entries

8. Explain all blocks of the system? Data download: Data ingest

Contar los bloques empleados en cada parte – como junto las piezas para construir el sistema – porque uso las piezas que uso

Hablar de implementaciones como que librerías, que scripts etc – no es muy necesario, mas importante la idea

5. REGULATORY FRAMEWORK

Not essential for this investigative project because we do nott go to market.

Habria que hablar de las normas, pq si alguien lo usa es para ir a mercado

I need to talk about laws, what data is permissible to use and how. This would be essential for a project that does indeed go to market.

For us, its not that relevant.

Software & Licenses: Visual Studio Code - general coding needs

Python - code platform

Scikit-learn - library for ml

Pandas - data management

Seaborn - data visualization

NumPy - number manipulation

LaTeX - open source language used to create the final project document

Overleaf - closed source platform used to compile LaTeX code

macOS / Windows - closed source platforms across where the project was developed

6. SOCIO-ECONOMIC ENVIRONMENT

6.1. Socio-economic Impact

Why is this relevant?

- Energy Prediction - Shutdowns - Price sensitivity

6.2. Project plan

Gantt diagram representing time spent - Tiempos de diseño e innovaacion. Amortizar nuestras horas de trabajo mias y emilio, ordenador y poco mas

6.3. Budget

Price breakdown of the investigation

7. CONCLUSIONS

7.1. Recap of the project

Executive review of the project

7.2. Revisit the objectives

Predicting the prices in an accurate manner using Machine Learning algorithms and technical analysis

7.3. Future work

Less error?

Menos piradas

BIBLIOGRAPHY

[1] OMIE, *Precios horarios del mercado diario en españa*, https://www.omie.es/es/file-access-list?parents=/Mercado%20Diario/1.%20Precios&dir=Precios%20horarios%20del%20mercado%20diario%20en%20EspaÃśa&realdir=marginalpdbc, Accessed on May 29, 2025, Jun. 2024.

[2] OMIE, *Modelo de ficheros para la distribución pública de información del mercado de electricidad 1.35*, https://www.omie.es/sites/default/files/2024-06/Formato_ficheros_inf_pub_135_0.pdf, Accessed on May 29, 2025, Jun. 2024.